

ROME Assignment Report
Course: LLM4ChipDesign
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Overview

This report documents the implementation and evaluation of a ROME-style hierarchical RTL generation workflow for Verilog design tasks. The notebook generates modules step-by-step using an LLM, verifies each generated module using ``iverilog`` and ``vvp``, and uses testbench feedback to refine incorrect designs through multiple iterations.

The workflow was applied to two required parts:

- Part I: provided mux hierarchy
- Part II: new hierarchical extension design

The experiments show that hierarchical decomposition makes it easier for the LLM to generate correct RTL incrementally, while simulation-based feedback helps fix syntax and functional errors.

System Architecture (ROME Hierarchical Loop)

The ROME workflow follows a hierarchical generate-test-refine process:

- Define a list of submodules from simplest to top-level
- Generate the first module from its interface specification
- Reuse previously generated modules while generating higher-level modules
- Write the generated Verilog into the module folders
- Compile each module together with its self-checking testbench using ``iverilog``.
- Simulate using ``vvp``.
- If compilation or simulation fails, feed the error output back into the next iteration
- Repeat until the module passes or the iteration budget is exhausted

This structure encourages module reuse and enforces hierarchical design instead of monolithic generation.

Model and Configuration

For this submission, the LLM was accessed through NVIDIA's OpenAI-compatible API.

Base URL: ``https://integrate.api.nvidia.com/v1``

Model family: NVIDIA

Model used: `nvidia/llama-3.1-nemotron-ultra-253b-v1`

API style: OpenAI-compatible chat completions

If a different NVIDIA model was used during the final run, replace the model name above with the exact one used in the notebook.

All designs were verified using:

```
```bash
iverilog
vvp
```
```

Key Features of Implementation

- Hierarchical generation of Verilog modules
- Reuse of previously generated submodules in later prompts
- Self-checking testbenches for every module
- Automatic iteration logging with `log\_iter\_\*.txt`
- Compiler/simulator feedback-driven correction loop
- Verification of both provided and extension hierarchies

Part I - Provided Mux Hierarchy

The provided demo hierarchy consisted of:

- `mux2to1`
- `mux4to1`
- `mux8to1`

Hierarchical decomposition

- `mux2to1` is the base submodule
- `mux4to1` is built using `mux2to1`
- `mux8to1` is built using `mux4to1` and `mux2to1`

Interfaces

```
```verilog
module mux2to1(input wire in1, input wire in2, input wire select, output wire out);
module mux4to1(input [1:0] sel, input [3:0] in, output wire out);
module mux8to1(input [2:0] sel, input [7:0] in, output wire out);
```
```

Verification

Each mux module was verified using a self-checking testbench. The testbench printed `passed!` on success, which was used by the notebook feedback loop to detect completion.

Final result

All mux hierarchy modules passed their testbenches successfully.

Part II - Extension Design: Ripple-Carry Adder Hierarchy

For the extension task, a ripple-carry adder hierarchy was implemented.

Module list

- `half\_adder`
- `full\_adder`
- `adder4`
- `adder8`

Hierarchical decomposition

- `half\_adder` is the base module
- `full\_adder` instantiates two `half\_adder` modules
- `adder4` instantiates four `full\_adder` modules
- `adder8` instantiates two `adder4` modules

This satisfies the assignment requirement that the top module instantiate at least two previously generated submodules.

Interfaces

```
```verilog
module half_adder(input wire a, input wire b, output wire sum, output wire carry);
module full_adder(input wire a, input wire b, input wire cin, output wire sum, output wire
cout);
module adder4(input wire [3:0] a, input wire [3:0] b, input wire cin, output wire [3:0] sum,
output wire cout);
module adder8(input wire [7:0] a, input wire [7:0] b, input wire cin, output wire [7:0] sum,
output wire cout);
```
```

Verification

Each module had a self-checking testbench:

- `half\_addertb.v`
- `full\_addertb.v`
- `adder4tb.v`
- `adder8tb.v`

Each testbench printed `passed!` when the design was correct.

Final result

All extension hierarchy modules passed their testbenches successfully.

Representative Prompts

Top module prompt

Generate an 8-bit ripple-carry adder hierarchically in Verilog using previously generated submodules half_adder, full_adder, and adder4.

Representative submodule prompt

Use the previously generated half_adder to build a full_adder.

Part III - Iterative Debugging Example

The iterative debugging process was documented for `adder8`.

Iteration 0: Compilation Failure

Issue observed:

- Syntax error in the generated `adder8` module

Evidence:

```
```text
```

```
HW3_ROME/adder8/adder8.v:50: syntax error
```

```
HW3_ROME/adder8/adder8.v:50: error: Invalid module instantiation
```

```
```
```

Fix applied:

- Corrected the malformed instantiation syntax in the `adder4` submodule connection

Outcome:

- The design compiled, but simulation still failed

Iteration 1: Functional Failure

Issue observed:

- Simulation produced unknown values

Evidence:

```
```text
```

```
adder8 mismatch: a=3c b=52 cin=0 expected_sum=8e expected_cout=0 got_sum=xx
got_cout=x
```
```

Fix applied:

- Refined the hierarchical response so the generated file included the required dependent submodule context correctly

Outcome:

- Unknown values were removed, but one functional mismatch remained

Iteration 2: Carry-Chain Error

Issue observed:

- Incorrect sum on a later test vector

Evidence:

```
```text
```

```
adder8 mismatch: a=4d b=6f cin=1 expected_sum=bd expected_cout=0 got_sum=ad
got_cout=0
```
```

Fix applied:

- Corrected the hierarchical carry propagation by ensuring the top module properly connected the low-nibble carry to the high-nibble stage

Outcome:

- Final design passed all test cases

Iteration 3: Final Passing Run

Result:

- `adder8` passed the self-checking testbench successfully

Final Verification Commands

The final designs were verified using commands of the following form:

```
```bash
iverilog -o mux2to1 mux2to1/mux2to1.v mux2to1/mux2to1tb.v
vvp mux2to1

iverilog -o mux4to1 mux4to1/mux4to1.v mux4to1/mux4to1tb.v
vvp mux4to1

iverilog -o mux8to1 mux8to1/mux8to1.v mux8to1/mux8to1tb.v
vvp mux8to1

iverilog -o half_adder half_adder/half_adder.v half_adder/half_addertb.v
vvp half_adder

iverilog -o full_adder full_adder/full_adder.v full_adder/full_addertb.v
vvp full_adder

iverilog -o adder4 adder4/adder4.v adder4/adder4tb.v
vvp adder4

iverilog -o adder8 adder8/adder8.v adder8/adder8tb.v
vvp adder8
```
```

Conclusion

The ROME hierarchical generation workflow successfully produced correct synthesizable Verilog for both the provided mux hierarchy and the new ripple-carry adder hierarchy. The decomposition strategy made generation more manageable by building larger designs from previously verified submodules.

The results show that:

- Hierarchical prompting improves RTL generation quality
- Self-checking simulation is essential for validating LLM-generated hardware
- Iterative feedback helps correct both syntax and logic errors
- Decomposition is especially useful for structured digital designs

All final submitted designs passed their corresponding testbenches successfully.